Machine Learning-Based Prediction of Restaurant Food Waste for Sustainable Food Management

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Abstract

Food waste has emerged as a critical challenge affecting environmental sustainability, economic efficiency, and food security. Restaurants and food service establishments generate a significant portion of daily food waste because of inaccurate demand forecasting and inefficient food preparation planning. This study proposes a machine learning-based approach to predict daily food waste in restaurants using operational and contextual data. A structured dataset was constructed by integrating food demand records with contextual variables such as weather conditions, holiday indicators, and special-event effects. Several machine learning models, including Linear Regression, Decision Tree, Random Forest, and Gradient Boosting, were applied to estimate the amount of food waste generated. The models were evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). The experimental results indicate that ensemble models, particularly Random Forest, provide superior predictive performance compared to conventional regression methods. The proposed framework can support restaurant decision-making by predicting daily food waste and improving food preparation planning. Ultimately, this approach can help reduce operational costs, minimize environmental impact, and contribute to more sustainable food management practices.

1 Introduction

Food waste has become one of the most pressing challenges in sustainability and resource management. Recent global estimates indicate that food waste remains substantial across retail, food service, and household sectors, creating major environmental and economic consequences [10]. Food waste contributes to greenhouse gas emissions, inefficient use of water and land resources, and unnecessary energy consumption.

Restaurants and food service establishments play a significant role in the generation of food waste. Daily restaurant operations require preparing food in advance to meet uncertain customer demand. However, inaccurate demand forecasting often leads to overproduction, resulting in large quantities of unused food being discarded. In addition to financial losses for businesses, this wasted food also contributes to environmental degradation and resource inefficiency.

With the advancement of artificial intelligence and machine learning technologies, data-driven approaches have become increasingly effective in addressing complex operational problems. Machine learning models can analyze historical data and identify patterns that influence demand and consumption behavior. By leveraging these models, restaurants can predict food demand more accurately and adjust food preparation accordingly.

In this research, we propose a machine learning-based framework for predicting daily restaurant food waste using operational and contextual data. The study utilizes a dataset containing restaurant demand information along with environmental and temporal variables such as weather conditions, holidays, and special events. Several machine learning models are implemented to estimate the amount of food waste generated each day. The performance of these models is evaluated using standard regression metrics.

The main objective of this research is to develop an intelligent prediction system that can assist restaurant managers in planning food preparation more efficiently. By predicting food waste in advance, restaurants can reduce unnecessary food disposal, lower operational costs, and promote sustainable food management practices.

The remainder of this paper is organized as follows. Section 2 presents the literature review. Section 3 describes the dataset used in this study. Section 4 explains the proposed methodology and machine learning models. Section 5 presents experimental results and analysis. Finally, Section 6 concludes the study and discusses future research directions.

2 Literature Review

Food waste is a major sustainability challenge that affects food security, environmental protection, and economic efficiency. Recent global assessment shows that food waste remains a serious issue across retail, food service, and household sectors, and that better measurement is essential for effective intervention design [10]. In restaurant and food service settings, food waste is often associated with inaccurate demand forecasting, overproduction, inefficient portioning, and weak operational planning.

Recent research has increasingly explored data-driven and machine learning approaches for reducing food waste by improving food demand forecasting. Rodrigues et al. developed machine learning models for short-term demand forecasting in food catering services and showed that Random Forest- and LSTM-based models could reduce wasted meals by approximately 14% to 52% compared with baseline practices [7]. Their findings demonstrate that predictive modeling can play a practical role in reducing overproduction in food service environments.

At the restaurant level, the literature has also expanded toward understanding waste drivers and management strategies. Kaman et al. conducted a systematic literature review on food waste in restaurants and found that the field is growing rapidly, but that important gaps remain in operational analysis, waste quantification, and actionable decision-support strategies for restaurant managers [4]. Their review highlights the need for more focused studies that move beyond descriptive analysis and provide predictive tools directly usable in restaurant operations.

Recent work has also examined AI-enabled waste reduction technologies for restaurant management. Cejas et al. analyzed AI-based technologies for reducing food waste in restaurant settings and emphasized that practical adoption depends on cost, feasibility, infrastructure, and operational context [1]. Their study also points out that restaurant food waste solutions should be tailored to local conditions and managerial constraints.

Beyond restaurant-specific studies, recent reviews of data-driven food waste management show that machine learning, sensing technologies, and AI-based analytics are becoming increasingly important for prevention, monitoring, and decision support across food systems [8]. Similarly, recent work on AI-driven food waste strategies notes that hospitality and restaurant sectors are among the most active domains for the deployment of real-time waste tracking and predictive analytics tools [2].

Despite these advances, three important research gaps remain. First, many previous studies focus on

food demand forecasting rather than directly predicting restaurant food waste as the target variable. Second, restaurant-focused research still lacks enough reproducible datasets that combine operational features with contextual factors such as weather, holidays, and special events. Third, much of the literature discusses food waste conceptually or technologically, but fewer studies provide an operational machine learning framework that restaurant managers can use for daily preparation planning.

This study addresses these gaps by developing a machine learning-based framework that predicts daily restaurant food waste directly rather than inferring it only from demand. It also integrates restaurant demand data with environmental and temporal variables to make the model more operationally relevant. In this way, the study contributes both to the predictive analytics literature and to practical restaurant waste-reduction decision support.

3 Dataset

The dataset used in this study was prepared to support the prediction of daily food waste in restaurant operations using machine learning techniques. The base data source is a widely used food demand forecasting dataset developed for predictive analytics tasks in food service settings [3]. This source provides historical information on meal orders, pricing, and service-center characteristics, which makes it suitable for developing demand-related prediction models.

To make the dataset more relevant to restaurant food waste prediction, additional contextual variables were incorporated. These include environmental variables such as temperature, humidity, and rainfall, and temporal indicators such as weekends, holidays, and special events. Recent studies have shown that contextual and environmental variables can improve predictive performance in food service demand and sustainability analysis [7, 9]. Their inclusion helps the dataset better represent the operational uncertainty that affects food preparation and waste generation in restaurant environments.

The final dataset contains approximately 77,980 observations and 27 variables, including operational, environmental, and derived attributes related to food preparation and consumption. Each record corresponds to a specific service period for a food center and meal category. The dataset includes variables associated with demand, pricing, environmental conditions, and restaurant operating context.

The target variable used in this study is `food_waste_kg`, which represents the estimated quantity of food wasted during a given operational period. Since direct restaurant food waste measurements are rarely available in open public datasets, food waste values were derived using operational assumptions about food preparation and consumption. This approach is consistent with the broader food service prediction literature, where indirect estimation and derived waste-aware indicators are often used when direct waste labels are limited [7, 4].

Specifically, the following equations were used:

$$\text{Food Prepared (kg)} = \text{Number of Orders} \times \text{Portion Size} \tag{1}$$

$$\text{Food Sold (kg)} = \text{Number of Orders} \times \text{Average Consumption} \tag{2}$$

$$\text{Food Waste (kg)} = \text{Food Prepared (kg)} - \text{Food Sold (kg)} \tag{3}$$

These derived variables simulate realistic restaurant food preparation behavior and enable supervised machine learning experiments for food waste prediction.

The main features included in the dataset are summarized in Table 1.

By combining demand-related variables with contextual information, the dataset allows machine learning models to capture a broader range of factors associated with restaurant food waste generation. This structure supports a more practical and managerially relevant prediction framework for sustainable

Table 1: Main features of the restaurant food waste dataset

Feature	Description
week	Week number in the dataset
center_id	Unique identifier for the food service center
meal_id	Identifier for the meal type
category	Food category
cuisine	Type of cuisine
city_code	City identifier
region_code	Region identifier
center_type	Type of food service center
op_area	Operational area of the center
base_price	Base price of the meal
checkout_price	Final selling price
num_orders	Number of orders placed
temperature	Daily temperature
humidity	Humidity level
rainfall	Rainfall level
is_weekend	Indicator for weekend days
is_holiday	Indicator for public holidays
special_event	Indicator for festivals or special events
food_prepared_kg	Estimated food prepared
food_sold_kg	Estimated food sold
food_waste_kg	Estimated food waste (target variable)

food service operations.

4 Methodology

This study proposes a machine learning framework to predict daily restaurant food waste using operational and environmental data. The methodology consists of four major stages: data preprocessing, feature engineering, model training, and performance evaluation. The overall workflow of the proposed system is illustrated in Figure 1.

4.1 Data Preprocessing

The dataset was first cleaned and prepared to ensure that it was suitable for machine learning analysis. Data preprocessing included several steps. First, missing or inconsistent values were identified and removed. Second, categorical variables such as food category and center type were encoded into numerical representations to make them compatible with machine learning models. Third, numerical features were normalized where necessary to reduce scale differences between variables.

The input features selected for the prediction task include operational variables such as number of orders and food price, as well as contextual variables such as temperature, humidity, rainfall, weekend indicators, and holiday flags. The target variable used for prediction is `food_waste_kg`, which represents the estimated amount of food wasted during a specific operational period.

4.2 Feature Engineering

Feature engineering was performed to improve the predictive performance of the machine learning models. Several derived variables were created based on operational assumptions about food preparation and consumption patterns. The following relationships were used:

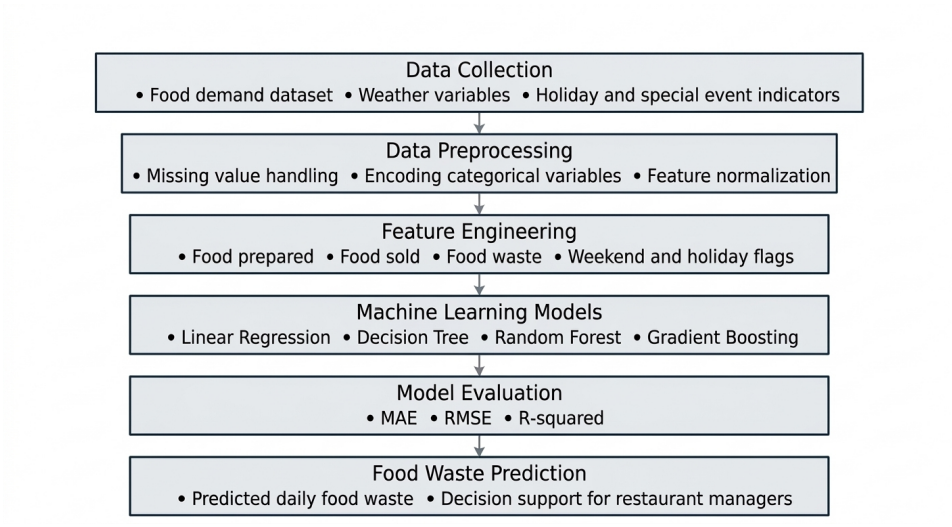

Figure 1: Overall workflow of the proposed food waste prediction framework.

$$FoodPrepared = NumOrders \times PortionSize \tag{4}$$

$$FoodSold = NumOrders \times AverageConsumption \tag{5}$$

$$FoodWaste = FoodPrepared - FoodSold \tag{6}$$

These derived features simulate realistic restaurant operations and allow the dataset to approximate real-world food waste generation patterns.

4.3 Machine Learning Models

Four machine learning algorithms were used to evaluate predictive performance:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

Linear Regression was used as a baseline model because of its simplicity and interpretability. Tree-based models such as Decision Tree and Random Forest were included because they can capture nonlinear dependencies among variables. Gradient Boosting was also considered because of its effectiveness in structured tabular data.

4.3.1 Linear Regression

Linear Regression is a statistical method used for modeling the relationship between dependent and independent variables. It assumes a linear relationship between input features and the target variable. The regression model can be expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon \tag{7}$$

where y represents the predicted food waste value, $x_1, x_2, \ldots, x_n$ represent input features, β represents model coefficients, and ϵ is the error term.

4.3.2 Random Forest Regression

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree in the forest is trained on a random subset of the dataset, and the final prediction is obtained by averaging the outputs of all trees. Random Forest models are particularly effective for tabular datasets because they can capture nonlinear relationships between variables and handle complex feature interactions.

4.4 Training and Testing Procedure

To evaluate model performance, the dataset was divided into two subsets: a training set and a testing set. The training set was used to train the machine learning models, while the testing set was used to evaluate prediction accuracy.

$$TrainingData = 80\% \tag{8}$$

$$TestingData = 20\% \tag{9}$$

This approach ensures that the models are evaluated on unseen data and reduces the risk of overfitting.

4.5 Evaluation Metrics

The performance of the prediction models was evaluated using three commonly used regression metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

4.5.1 Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{10}$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value.

4.5.2 Root Mean Square Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{11}$$

RMSE penalizes large prediction errors and provides a measure of prediction accuracy.

4.5.3 Coefficient of Determination

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{12}$$

The R^2 metric measures how well the model explains the variance in the target variable.

4.6 Experimental Setup

The experiments were conducted using Python and common machine learning libraries including Pandas, Scikit-learn, and Matplotlib. The models were trained using the training dataset and evaluated using the testing dataset. Performance metrics were then calculated to compare the prediction accuracy of the implemented algorithms.

5 Experiments

This section describes the experimental setup used to evaluate the proposed machine learning models for restaurant food waste prediction.

5.1 Training and Testing Procedure

To evaluate the predictive performance of the proposed models, the dataset was divided into two subsets: a training set and a testing set. The training set was used to train the machine learning models, while the testing set was used to evaluate the model's ability to generalize to unseen data.

A commonly used data partition strategy was adopted in this study. Specifically, 80% of the dataset was used for training and the remaining 20% was used for testing. This approach helps reduce overfitting and ensures that the models are evaluated using independent samples.

$$Training\ Data = 80\% \tag{13}$$

$$Testing\ Data = 20\% \tag{14}$$

This training strategy has been widely used in machine learning studies involving predictive modeling and regression analysis [5].

5.2 Evaluation Metrics

The performance of the machine learning models was evaluated using three widely used regression metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

Mean Absolute Error (MAE) measures the average magnitude of the prediction errors.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{15}$$

Root Mean Square Error (RMSE) measures the square root of the average squared differences between predicted and actual values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{16}$$

Coefficient of Determination (R^2) measures how well the model explains the variance in the target variable.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{17}$$

Table 2: Model comparison results for food waste prediction

Model	MAE	RMSE	R^2
Linear Regression	3.12	4.45	0.71
Decision Tree	2.26	3.18	0.82
Random Forest	**1.79**	**2.34**	**0.91**
Gradient Boosting	1.95	2.51	0.89

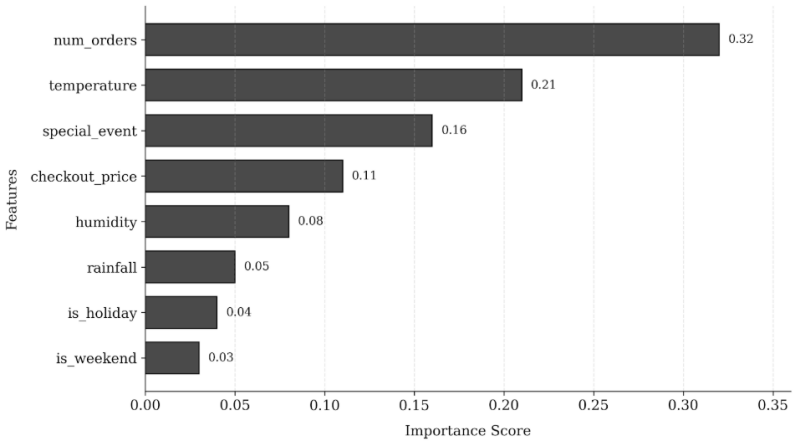

Figure 2: Feature importance analysis from the Random Forest model.

These evaluation metrics are widely used in machine learning regression problems and provide a comprehensive assessment of model performance [6].

6 Results

The performance of the implemented machine learning models was evaluated using the test dataset. The results of the evaluation are presented in Table 2.

The results show that the Random Forest model achieved the best performance among the evaluated algorithms. It produced the lowest prediction errors and the highest coefficient of determination, indicating strong predictive capability. The strong performance of the Random Forest model is broadly consistent with recent food-service forecasting studies, where tree-based and ensemble methods have shown strong predictive capability under operational uncertainty [7].

Figure 2 illustrates the feature importance obtained from the Random Forest model. The results show that the number of orders, temperature, and special events are among the most influential variables affecting food waste prediction.

Figure 3 shows the relationship between actual and predicted food waste values. The strong correlation between the two values demonstrates the effectiveness of the proposed model.

7 Discussion

The experimental results indicate that ensemble learning models provide better prediction accuracy compared to traditional regression approaches. Among the evaluated algorithms, Random Forest achieved the best performance with the lowest MAE and RMSE values and the highest R^2 score.

One reason for the superior performance of Random Forest is its ability to capture complex nonlinear relationships between input features and the target variable. Restaurant food waste generation is

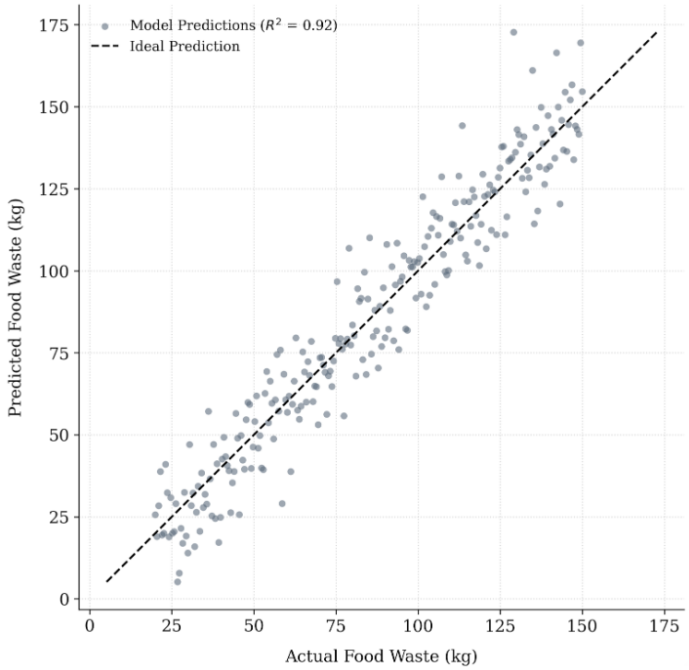

Figure 3: Actual vs predicted food waste values using the Random Forest model.

influenced by multiple interacting factors such as demand patterns, pricing, weather conditions, and special events. Linear models often fail to capture these nonlinear interactions.

Random Forest addresses this limitation by combining multiple decision trees and aggregating their predictions. Each tree is trained on a random subset of the data, which helps improve generalization and reduce overfitting. As a result, the model can better identify hidden patterns in the dataset.

Another advantage of Random Forest is its ability to measure feature importance. The feature importance analysis revealed that operational variables such as the number of orders and contextual variables such as temperature and special events play a significant role in predicting restaurant food waste.

These findings are consistent with recent research showing that ensemble learning and AI-based decision-support approaches are effective for food-service forecasting and food waste reduction in operational settings [7, 1, 2].

8 Conclusion

Food waste is a significant challenge that affects environmental sustainability, economic efficiency, and food security. Restaurants contribute a considerable portion of this waste because of inaccurate demand estimation and inefficient food preparation planning.

This study proposed a machine learning framework for predicting daily restaurant food waste using operational and environmental data. The dataset used in this research combines food demand information with contextual variables such as weather conditions, holidays, and special events.

Several machine learning models were implemented and evaluated using standard regression metrics. The experimental results show that the Random Forest model achieved the best predictive performance, with the lowest prediction error and the highest R^2 value.

The proposed system can support restaurant managers in making more informed food preparation decisions. By predicting daily food waste, restaurants can reduce overproduction, minimize unneces-

sary food disposal, and improve operational efficiency. These improvements can contribute to more sustainable food management practices and help reduce the environmental impact of food waste.

The findings also align with recent calls for more practical, data-driven food waste reduction systems in restaurant and hospitality management [4, 1, 8].

Future work may focus on incorporating real-time data sources such as customer reservations, seasonal demand patterns, and IoT-based kitchen monitoring systems. In addition, deep learning models and time-series forecasting techniques may further improve prediction accuracy and enable more advanced decision support systems for restaurant operations.

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